

Answer Key — Chemistry Grade 10

Sub-Strand 3.1: Acids and Bases • Lesson 4: Differences Between Acids and Bases Using Commercial Indicators

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	A	A	B	A	A	D

USE AFTER: PREDICT PHASE — Think-Pair and cold-call poll of the pre-lab prediction table

Q1. Two partners disagree about what colour methyl orange will turn in baking soda solution. Before testing, what is the best way to settle the disagreement?

- A. Carry out the test and record the colour actually seen**
- B. Ask the teacher to say which prediction is right
- C. Vote and go with the answer most students choose
- D. Let the student who is more confident decide

Correct answer: A. In science, disagreements between predictions are settled by evidence from the experiment, not by confidence, authority or votes.

USE AFTER: OBSERVE PHASE — Experiment 1 procedure, Step 1: litmus strips with Test Solution 1 (lemon juice)

Q2. In Experiment 1, a student puts 2-3 drops of freshly squeezed lemon juice on a strip of blue litmus paper. What colour does the paper turn?

- A. Red**
- B. Blue (no change)
- C. Pink
- D. Colourless

Correct answer: A. Lemon juice is an acid, and acids turn blue litmus paper red.

USE AFTER: OBSERVE PHASE — Experiment 1: circulation and 'check for honest recording' while groups fill the results table

Q3. A group predicted that cow's milk would be strongly basic. Their litmus and indicator results do not match this prediction. What should the recorders do?

- A. Write down the colours exactly as seen, even if they differ from the prediction**
- B. Leave the milk row blank because the result was unexpected
- C. Copy the results from another group's table
- D. Change the results in the table to match the prediction

Correct answer: A. Honest, accurate recording is essential. An unexpected result is useful data and must never be altered to fit a prediction.

USE AFTER: EXPLAIN PHASE — Consensus Results Table and pattern rules for each indicator (Explain phase)

Q4. Liquid soap solution turns red litmus paper blue and turns phenolphthalein pink. How should it be classified, and why?

- A. Acid, because it changes the colour of litmus paper
- B. Base, because red litmus turns blue and phenolphthalein turns pink**
- C. Neutral, because it does not make methyl orange red
- D. Acid, because phenolphthalein changes colour

Correct answer: B. Bases turn red litmus blue and phenolphthalein pink or magenta. Both results point to a base.

USE AFTER: EXPLAIN PHASE — Pattern Statement: writing rules for acids, bases and neutral substances from the consensus table

Q5. A student tests tap water. Litmus paper shows no colour change, phenolphthalein stays colourless and methyl orange is orange-yellow. What is the best conclusion?

- A. It is neutral, because none of the indicators show an acid or base colour**
- B. It cannot be classified because no indicator turned red
- C. It is a base, because methyl orange is yellowish
- D. It is an acid, because phenolphthalein is colourless

Correct answer: A. Neutral substances do not change litmus, leave phenolphthalein colourless and make methyl orange orange-yellow. Colourless phenolphthalein alone cannot separate acids from neutral liquids, so the other indicators must be used as well.

USE AFTER: DQB PHASE — DQB: sticky-note questions, clustering, and the SOLVED zone

Q6. After Experiment 1, students can tell whether a liquid is an acid, a base or neutral using indicators. Which question is still NOT answered by this experiment?

- A. How do acids react, for example do they corrode or react with other substances?**
- B. Is baking soda solution a base?
- C. What colour does phenolphthalein turn in a base?
- D. Is vinegar (siki) an acid?

Correct answer: A. Indicators only classify a liquid by colour change. They do not show what an acid will do chemically, and this question leads into the next lesson. The other three questions were answered by today's results.

USE AFTER: MODEL BUILDING PHASE — Anchor model revision: labelling each cup as ACID, BASE or NEUTRAL

Q7. A student revises a model by labelling the liquids tested in Experiment 1 using the evidence. Which set of labels is correct?

- A. Lemon juice: base; baking soda solution: acid; tap water: neutral
- B. Lemon juice: neutral; baking soda solution: base; tap water: acid
- C. Lemon juice: acid; baking soda solution: neutral; tap water: base
- D. Lemon juice: acid; baking soda solution: base; tap water: neutral**

Correct answer: D. Lemon juice is an acid (blue litmus turns red). Baking soda solution is a base (red litmus turns blue, phenolphthalein turns pink). Tap water is neutral (no change).